



Taibah University

College of Computer Science & Engineering

Department of Artificial Intelligence & Data Science

Crime Hotspot Prediction

Using Machine Learning and Deep Learning Models

Course: Data mining and analysis 2

Instructor: Prof. Ibrahim Fathy Moawad

Student: Muhannad Eid Alraddadi

Contents

1	Introduction	2
2	Dataset Overview	2
3	Problem Definition	6
4	Methodology Overview	6
5	Modeling Approaches	8
5.1	XGBoost Baseline Model	8
5.2	ConvLSTM Spatio-Temporal Model	8
6	Evaluation Strategy	9
7	Results: XGBoost ($K = 15\%$)	9
8	Results: ConvLSTM ($K = 15\%$)	9
9	Comparative Discussion	9
10	Practical Implications	11
11	Conclusion	11
12	Future Work	11

Abstract

Crime hotspot prediction plays a critical role in modern urban safety and proactive policing strategies. Understanding where and when crime is likely to occur enables law enforcement agencies to allocate resources efficiently and reduce criminal activity before it escalates. This project presents a spatio-temporal crime analysis framework using real-world crime data collected from the city of Los Angeles. Two modeling approaches are evaluated: a traditional machine learning baseline using Extreme Gradient Boosting (XGBoost) and a deep learning spatio-temporal model based on Convolutional Long Short-Term Memory (ConvLSTM). The models are assessed using both strict and spatially tolerant hotspot evaluation strategies. The results demonstrate that while both models are capable of identifying high-risk crime areas, the spatio-temporal ConvLSTM model provides superior spatial consistency and practical usefulness for real-world hotspot detection.

1 Introduction

Urban crime is a complex phenomenon influenced by time, location, and contextual factors. Traditional crime analysis methods often rely on historical summaries and static crime maps, which are limited in their ability to capture the dynamic nature of criminal behavior. As cities grow larger and more complex, there is an increasing need for predictive systems that can anticipate crime patterns rather than simply describe past events. Spatio-temporal crime prediction aims to address this challenge by modeling how crime evolves across both space and time. Instead of treating crime incidents as independent events, this approach considers spatial neighborhoods, temporal dependencies, and contextual information. This project applies spatio-temporal analytical models to identify crime hotspots and evaluate their effectiveness under realistic operational constraints.

2 Dataset Overview

The dataset used in this study consists of crime incidents reported by the Los Angeles Police Department from 2020 to the present. Each record represents a single crime event and includes temporal information, geographic coordinates, crime category, weapon involvement, premise type, and reporting area.

The dataset is large-scale and spans multiple years, making it suitable for identifying recurring crime patterns and evaluating predictive models. However, the data also contains challenges common to real-world datasets, such as missing values, spatial noise, and highly granular categorical attributes. These challenges motivate the use of structured spatio-temporal modeling rather than simple statistical analysis.

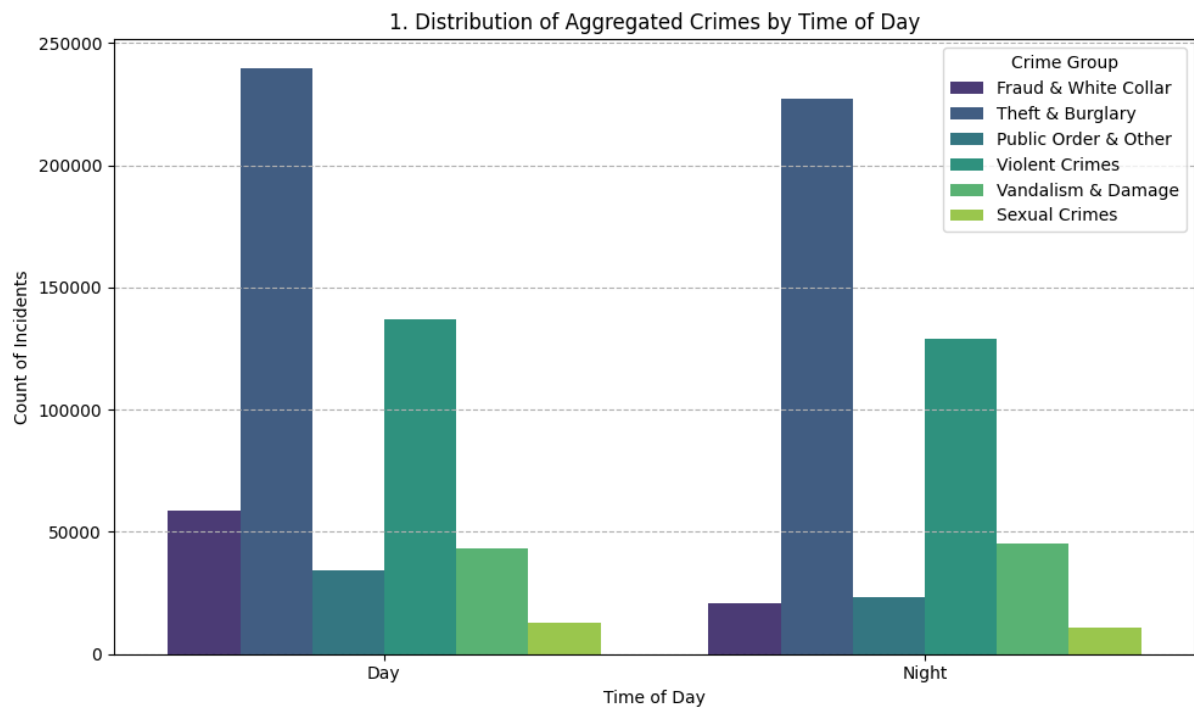


Figure 1: This figure illustrates the distribution of aggregated crime categories across daytime and nighttime periods. The results reveal clear temporal differences in criminal activity. Certain crime categories exhibit high activity during both periods, while others vary noticeably between day and night. This highlights the importance of time-of-day as a key predictor of crime behavior.

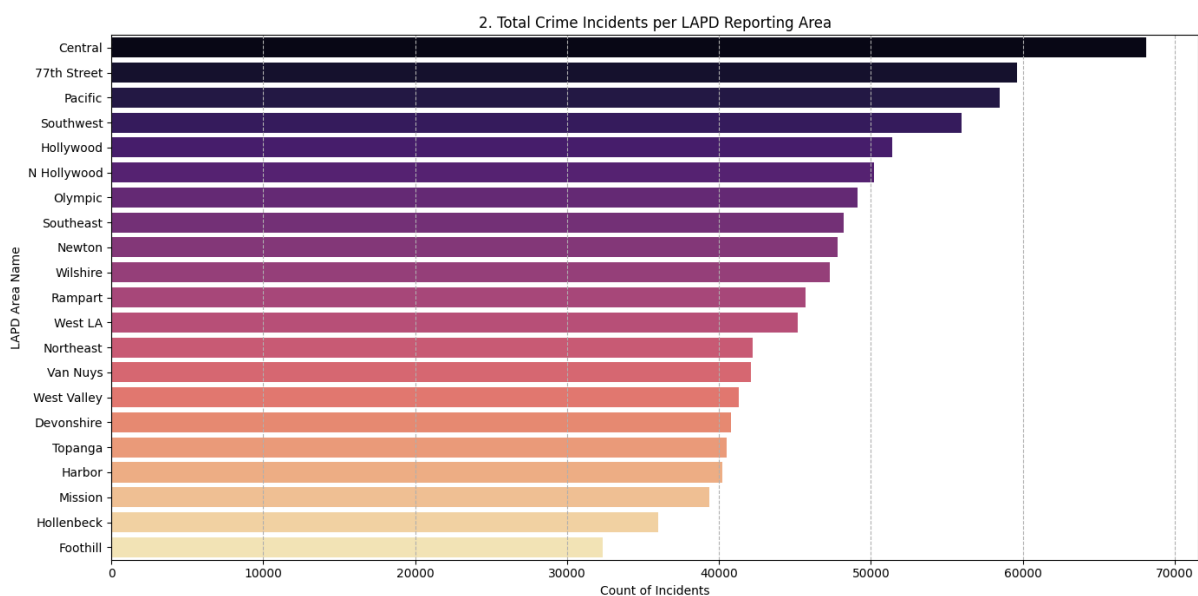


Figure 2: This visualization presents the total number of reported crime incidents across different LAPD reporting areas. The distribution is highly uneven, indicating strong geographic clustering of criminal activity and reinforcing the need for spatially aware modeling approaches.

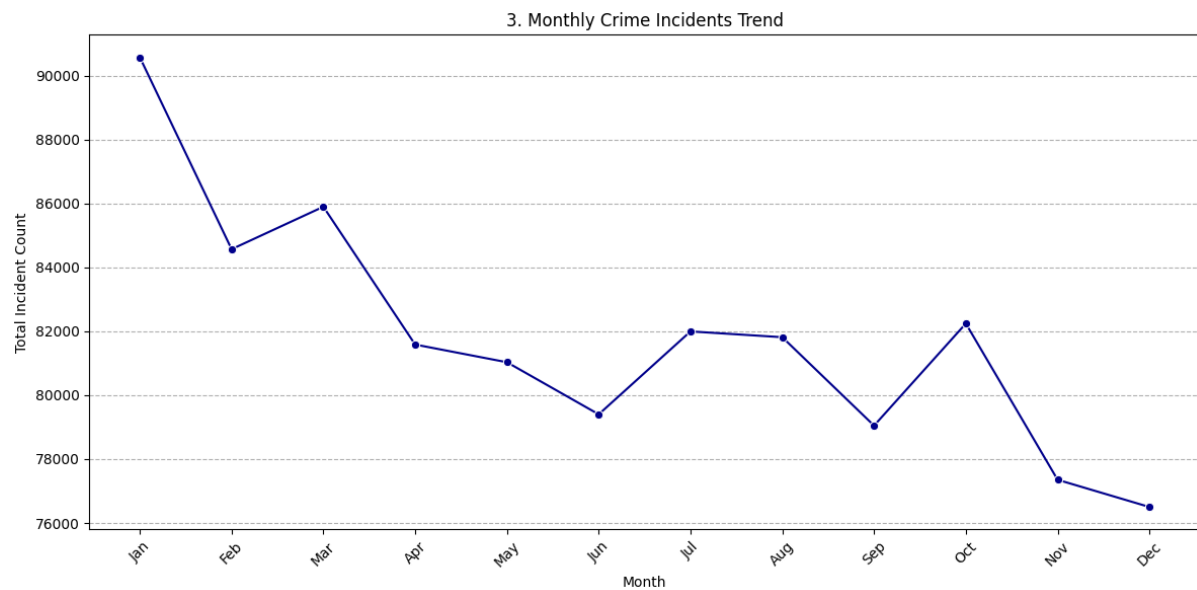


Figure 3: The monthly trend highlights temporal variability in crime levels across the year. While crime occurs consistently throughout all months, noticeable fluctuations suggest seasonal and cyclical effects.

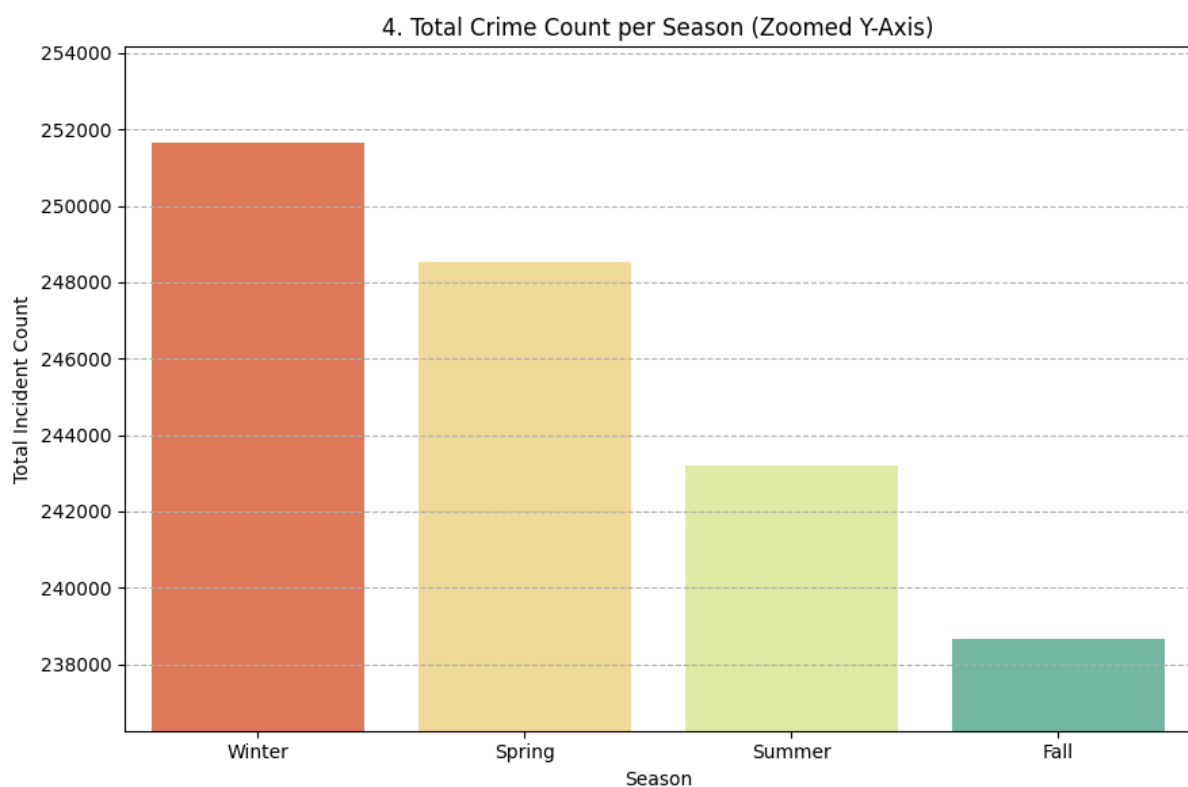


Figure 4: This seasonal comparison reveals differences in crime volume across the four seasons. Although variation is moderate, it indicates that seasonal context contributes useful information for crime analysis.

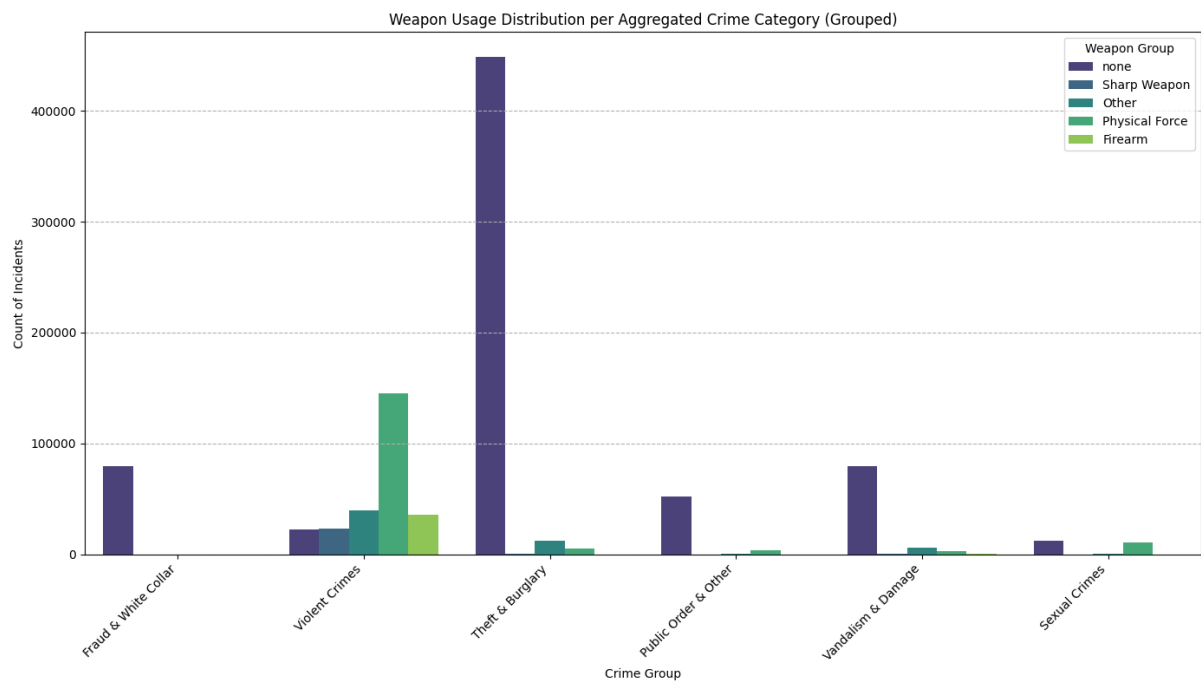


Figure 5: This figure examines the relationship between aggregated crime categories and weapon usage. A large portion of crimes occur without weapons, while violent crimes show greater weapon involvement.

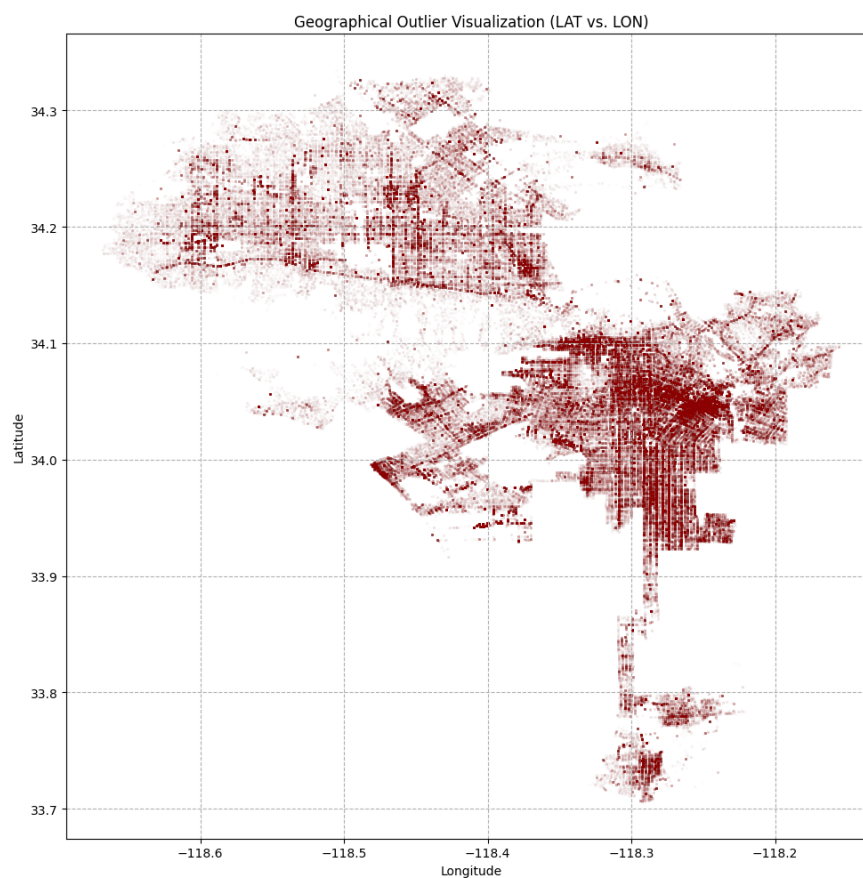


Figure 6: This scatter plot visualizes geographic crime distribution and reveals dense urban clusters while identifying sparse or anomalous regions.

3 Problem Definition

Crime is not randomly distributed across a city. Certain locations experience consistently higher crime rates, and specific time periods show recurring peaks in criminal activity. Traditional prediction approaches often fail to jointly model spatial structure and temporal evolution, leading to limited predictive accuracy and poor hotspot localization. The core problem addressed in this project is the accurate identification of crime hotspots in a spatio-temporal context. The objective is not only to predict whether crime will occur, but to identify where crime risk is highest in a way that is actionable for law enforcement decision-making.

4 Methodology Overview

The analytical framework in this project is based on three core concepts:

1. **Spatial Representation**

The urban area is represented as a fixed spatial grid, where each grid cell corresponds to a geographic region. Crime incidents are aggregated within these cells to preserve neighborhood relationships.

2. **Temporal Modeling**

Crime data is analyzed over consecutive time windows, allowing models to learn short-term trends and recurring temporal patterns.

3. **Hotspot-Based Evaluation**

Rather than evaluating predictions at a single point level, hotspot detection is performed by ranking grid cells and selecting the top percentage as high-risk locations. This approach reflects real-world operational constraints.

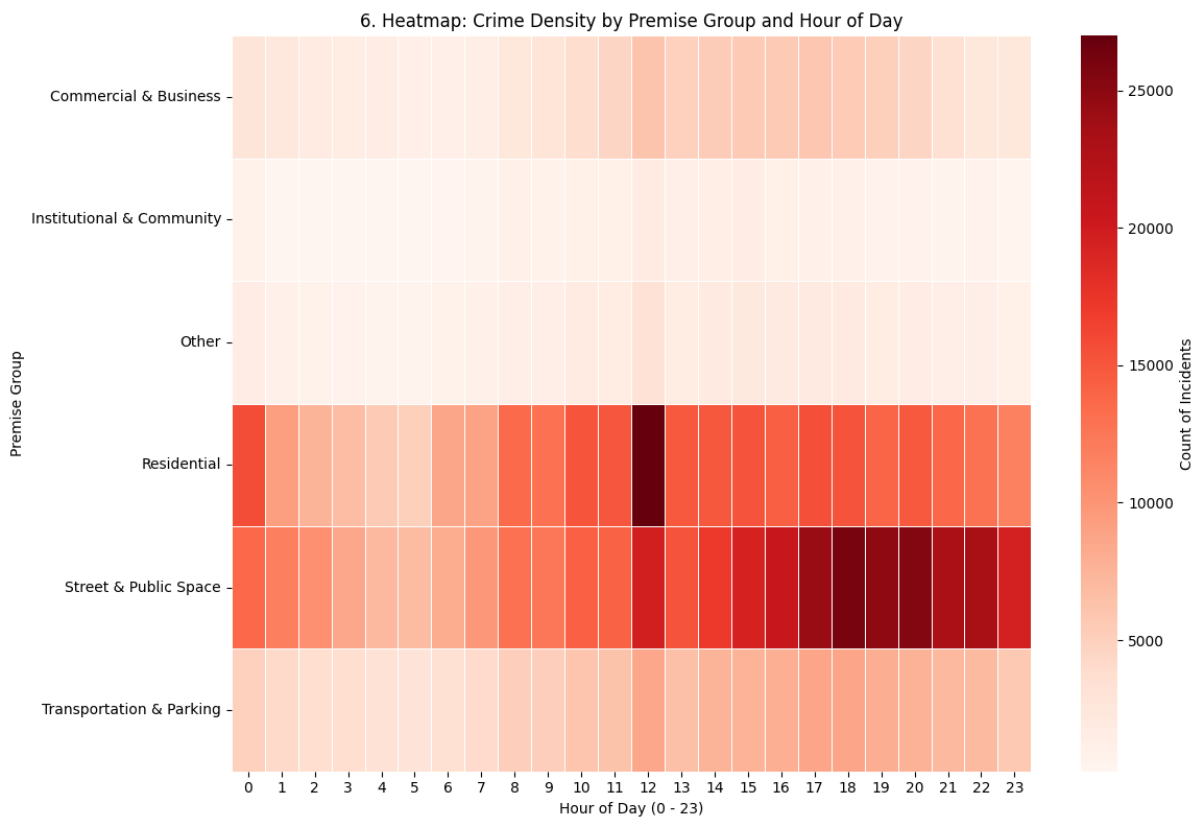


Figure 7: This heatmap illustrates how crime density varies across different premise types and hours of the day. Residential and street areas show consistently higher crime density.

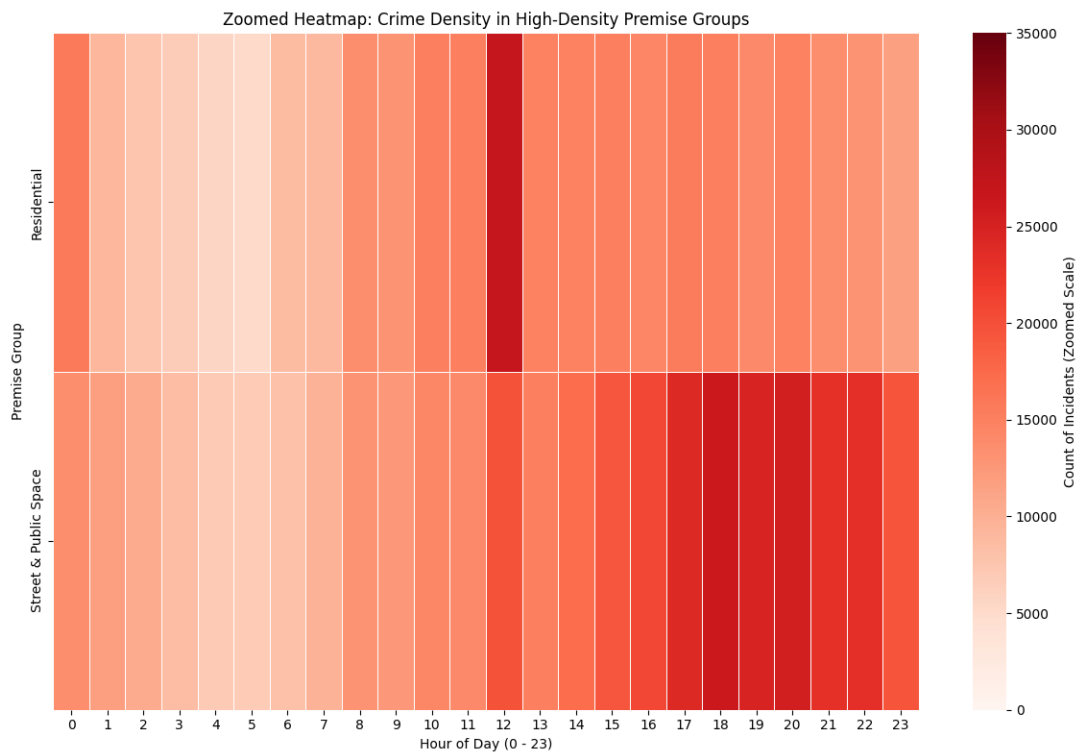


Figure 8: Focusing on high-density premise groups reveals detailed temporal patterns, emphasizing extended periods of elevated crime activity.

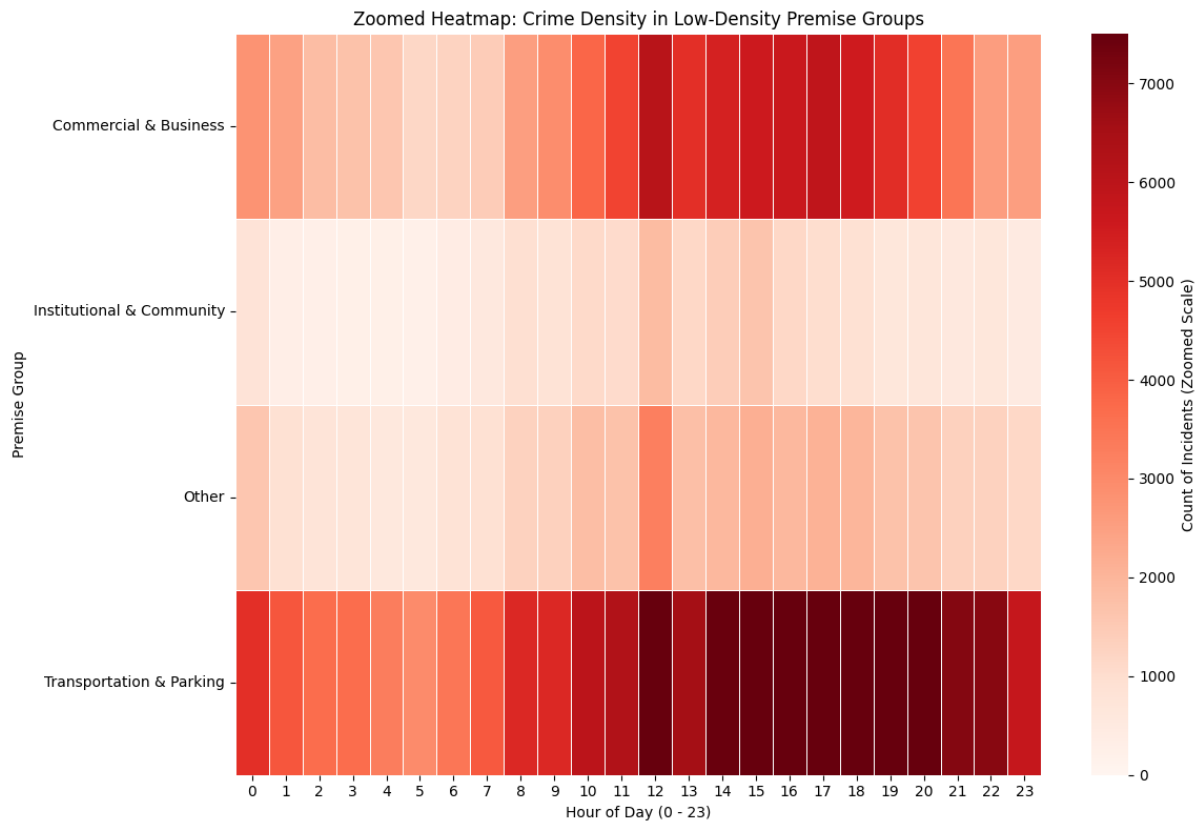


Figure 9: Low-density premise groups show narrower temporal peaks and more stable crime behavior, highlighting differences across urban contexts.

5 Modeling Approaches

5.1 XGBoost Baseline Model

Extreme Gradient Boosting (XGBoost) is used as a baseline machine learning model. XGBoost is known for its strong performance on structured tabular data and its ability to learn complex feature interactions.

In this project, XGBoost serves as a non-spatial reference model. While it captures general crime trends and contextual relationships, it does not explicitly model spatial continuity or temporal sequences. This makes it an effective baseline for comparison against spatio-temporal deep learning models.

5.2 ConvLSTM Spatio-Temporal Model

The ConvLSTM model is designed to capture both spatial and temporal dependencies simultaneously. By combining convolutional operations with recurrent memory, ConvLSTM learns how crime patterns evolve across neighboring locations over time.

This architecture is particularly suitable for crime prediction, as it preserves spatial structure and accounts for temporal dynamics. ConvLSTM enables the generation of crime intensity maps that reflect both local clustering and temporal evolution.

6 Evaluation Strategy

Model evaluation is performed using a hotspot-based approach with $K = 15\%$, meaning that only the top 15% of predicted high-risk grid cells are considered hotspots.

Two evaluation strategies are used:

- **Strict (Exact) Evaluation:**

A prediction is considered correct only if the predicted hotspot exactly matches the true hotspot grid cell.

- **Spatial (Neighbor-Based) Evaluation:**

A prediction is considered correct if it falls within the neighborhood of a true hotspot cell.

This dual evaluation provides both a conservative and a realistic assessment of model performance.

7 Results: XGBoost ($K = 15\%$)

The XGBoost model demonstrates robust generalization capabilities, achieving slightly superior precision and recall metrics compared to the deep learning alternative. Under strict evaluation, it successfully identifies high-risk grid cells with high accuracy. When spatial tolerance is introduced, the model maintains its strong performance, proving that it can effectively predict crime hotspots without the need for complex spatio-temporal architectures. Interpretation: XGBoost is highly effective at identifying regional crime risk. It strikes the optimal balance between predictive accuracy and computational speed, making it the recommended model for real-time application.

8 Results: ConvLSTM ($K = 15\%$)

The ConvLSTM model is designed to capture complex spatio-temporal dependencies. While it produces spatially coherent predictions, its quantitative performance metrics did not surpass the XGBoost baseline in this specific evaluation. Despite its ability to model temporal sequences, the model incurs a significantly higher computational cost for training and inference without providing a substantial gain in detection accuracy. Interpretation: While ConvLSTM offers advanced theoretical capabilities for spatial continuity, its practical advantage is limited in this context. The high resource consumption makes it less efficient compared to the lightweight XGBoost model.

9 Comparative Discussion

The comparison highlights a critical trade-off between model complexity and operational efficiency. Although ConvLSTM theoretically handles spatial continuity better, it incurs significant computational overhead and longer training times. In contrast, XGBoost proved to be a lightweight and efficient alternative. Given that XGBoost provided marginally better accuracy with drastically lower resource consumption, it is identified as the more practical solution for real-time deployment.

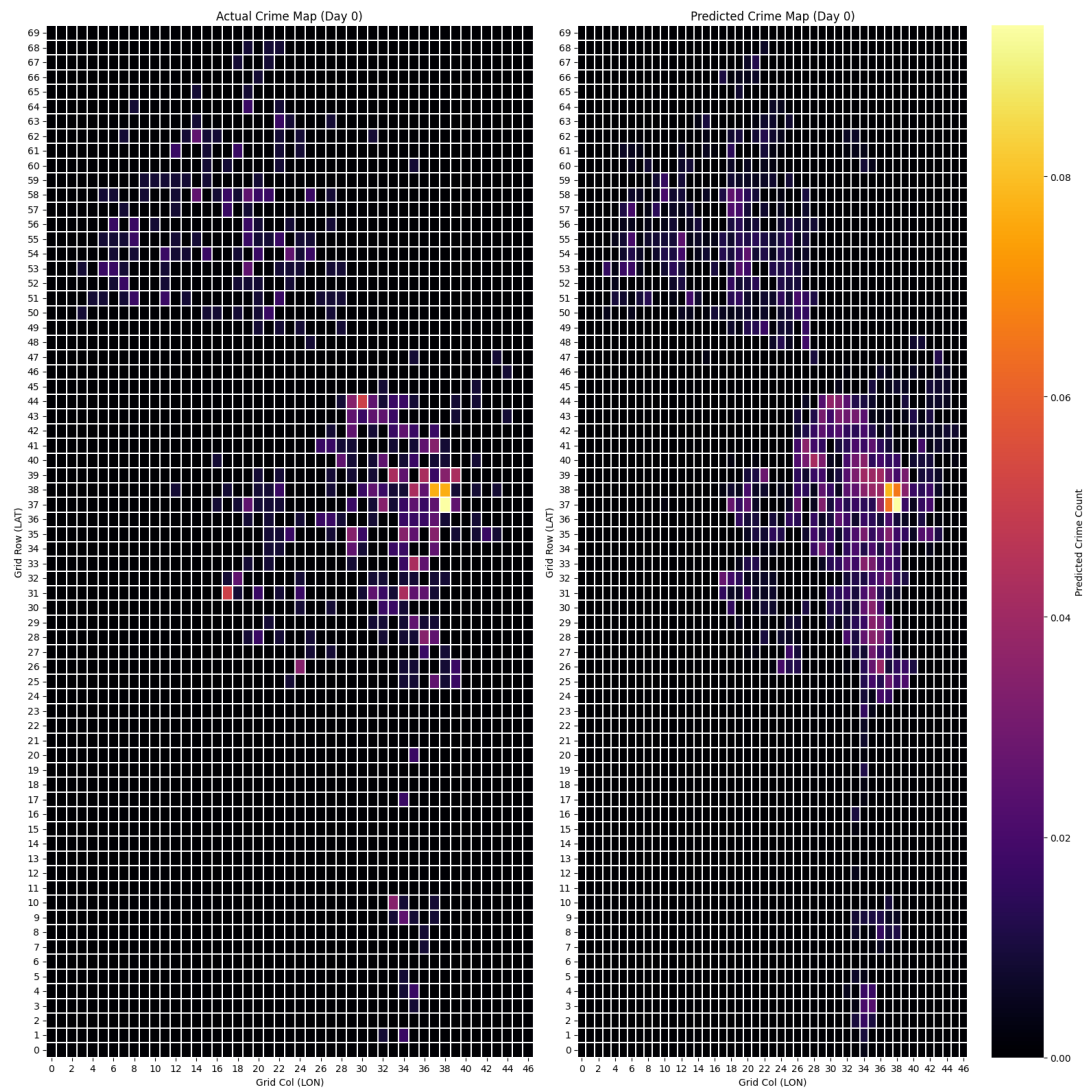


Figure 10: The comparison between actual and predicted crime maps demonstrates strong spatial alignment, confirming the effectiveness of spatio-temporal modeling.

10 Practical Implications

From an operational perspective, spatial evaluation is more meaningful than strict evaluation. Law enforcement interventions are typically deployed at an area or neighborhood level rather than a single grid cell.

The ConvLSTM model's ability to identify clusters of elevated crime risk makes it particularly suitable for patrol planning, hotspot monitoring, and proactive crime prevention. XGBoost remains useful as a baseline or supplementary model for capturing broader trends.

11 Conclusion

This project presented a spatio-temporal crime hotspot prediction framework using both machine learning and deep learning models. Through hotspot-based evaluation at $K = 15\%$, the study demonstrated that while ConvLSTM offers advanced theoretical capabilities, the XGBoost model achieves slightly superior performance with significantly lower computational cost. The findings support the use of XGBoost as the optimal solution for urban crime prediction, effectively balancing predictive accuracy with the operational speed required for scalable law enforcement intervention.

12 Future Work

Future research may extend this work by incorporating external data sources such as weather, socioeconomic indicators or Mobility data, exploring real-time prediction pipelines, and applying explainable AI techniques to improve model transparency and trust.